

# Jongmin Moon

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## RESEARCH KEYWORDS

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Computational Neuroscience · Systems Neuroscience · Vision · Decision Making · Neural Coding

## APPOINTMENTS

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2024 – present      **Postdoctoral Fellow**  
Center for Perceptual Systems  
The University of Texas at Austin, Austin, TX, USA

## EDUCATION

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2018 – 2024      **Ph.D.**, Biomedical Engineering  
Ulsan National Institute of Science and Technology, Ulsan, South Korea  
• Thesis: *Common computational principles underlying biases in visual perception, memory and confidence*

2012 – 2018      **B.S.**, Human Factors Engineering; System Design and Control Engineering  
Ulsan National Institute of Science and Technology, Ulsan, South Korea

## PUBLICATIONS (\*equal contribution; mentee collaborators)

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### Peer-Reviewed Journal Articles

1. Saftari, L. N.\*, **Moon, J.\***, & Kwon, O.-S. (2025). Environmental motion presented ahead of self-motion modulates heading direction estimation. *PLOS Computational Biology*, 21(10), e1013571.
2. **Moon, J.**, Tadin, D.\*, & Kwon, O.-S.\* (2023). A key role of orientation in the coding of visual motion direction. *Psychonomic Bulletin & Review*, 30, 564–574.
3. **Moon, J.**, & Kwon, O.-S. (2022). Attractive and repulsive effects of sensory history concurrently shape visual perception. *BMC Biology*, 20, 247.
4. **Moon, J.\***, Choe, S.\*, Lee, S., & Kwon, O.-S. (2019). Temporal dynamics of visual attention allocation. *Scientific Reports*, 9, 3664.

### Commentary Articles

1. **Moon, J.**, & Goris, R. L. T. (2026). Image space opens up for visual neuroscience. *Neuron*, 114, 2296–2297.
  - Preview of: Fu et al. (2026). Statistics of natural scenes shape contextual modulation in the visual cortex. *Neuron*, 114, 2473–2486.e9.

### Manuscripts Under Review or In Preparation

1. **Moon, J.**, Chung, D., Kim, H. J., Sul, S., & Kwon, O.-S. (under review). Rational confidence judgments in optimal and suboptimal perceptual estimations.
2. **Moon, J.**, Yoon, H., & Kwon, O.-S. (in preparation). A behavioral signature of efficient coding in working memory representations.

## CONFERENCE PRESENTATIONS

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1. **Moon, J.**, Boundy-Singer, Z. M., Charlton, J. A., & Goris, R. L. T. (2026, November 14–18). *How history manifests in the macaque cortical hierarchy*. [submitted]. Society for Neuroscience (SfN) Annual Meeting (Neuroscience 2026), Washington, DC, USA.
2. **Moon, J.**, Boundy-Singer, Z. M., Charlton, J. A., & Goris, R. L. T. (2026, May 15–19). *How history manifests in the macaque cortical hierarchy*. [Talk session]. Vision Sciences Society (VSS) Annual Meeting, St. Pete Beach, FL, USA.
3. Ranjan, A., **Moon, J.**, & Goris, R. L. T. (2026, May 15–19). *Confidence in perceptual estimates reveals that sensory noise is doubly stochastic*. [Poster session]. Vision Sciences Society (VSS) Annual Meeting, St. Pete Beach, FL, USA.
4. Gunniya Prakash, A. R., **Moon, J.**, & Goris, R. L. T. (2026, May 15–19). *Perceptual confidence seems blind to prior information*. [Poster session]. Vision Sciences Society (VSS) Annual Meeting, St. Pete Beach, FL, USA.
5. Yoon, H., **Moon, J.**, & Kwon, O.-S. (2025, May 16–20). *Attractive serial dependence occurs after motion repulsion*. [Poster session]. Vision Sciences Society (VSS) Annual Meeting, St. Pete Beach, FL, USA.
6. **Moon, J.**, Gunniya Prakash, A. R., & Goris, R. L. T. (2025, April 25–27). *Bayesian confidence paradox*. [Poster session]. Workshop on Natural Environments, Tasks, and Intelligence (NETI), Austin, TX, USA.
7. **Moon, J.**, Yoon, H., & Kwon, O.-S. (2024, May 17–22). *Probing bidirectional serial dependence in an N-back orientation estimation task*. [Talk session]. Vision Sciences Society (VSS) Annual Meeting, St. Pete Beach, FL, USA.
8. **Moon, J.**, Saftari, L. N., & Kwon, O.-S. (2023, August 24–27). *Environmental motion presented ahead of self-motion modulates the heading direction estimation*. [Poster session]. Conference on Cognitive Computational Neuroscience (CCN), Oxford, United Kingdom.
9. Saftari, L. N., **Moon, J.**, & Kwon, O.-S. (2023, May 19–24). *Synchronicity of visual and vestibular signals modulates the causal inference in heading direction estimation*. [Poster session]. Vision Sciences Society (VSS) Annual Meeting, St. Pete Beach, FL, USA.
10. **Moon, J.**, & Kwon, O.-S. (2022, May 13–18). *Confidence in perceptual estimation reflects behavioral variability, but not biases*. [Poster session]. Vision Sciences Society (VSS) Annual Meeting, St. Pete Beach, FL, USA.
11. **Moon, J.**, & Kwon, O.-S. (2020, June 19–24). *Motion perception is biased toward the orientation, not the direction, of the preceding motion*. [Poster session]. Vision Sciences Society (VSS) Annual Meeting, virtual.
12. **Moon, J.**, & Kwon, O.-S. (2019, May 17–22). *Additivity of attractive and repulsive sequential effects in motion direction estimation*. [Talk session]. Vision Sciences Society (VSS) Annual Meeting, St. Pete Beach, FL, USA.
13. Choe, S., **Moon, J.**, & Kwon, O.-S. (2018, May 18–23). *Temporal dynamics of visual attention allocation*. [Poster session]. Vision Sciences Society (VSS) Annual Meeting, St. Pete Beach, FL, USA.

## Domestic Conference Presentations

1. **Moon, J.**, Boundy-Singer, Z. M., Charlton, J. A., & Goris, R. L. T. (2026, September 6–8). *How history manifests in the macaque cortical hierarchy*. [submitted]. The 29th Annual Meeting of the Korean Society for Brain and Neural Sciences (KSBNS), Daejeon, South Korea.
2. Yoon, H., **Moon, J.**, & Kwon, O.-S. (2026, January 21–23). *Sequential effects of suppressed stimuli in motion direction perception*. [Poster session]. The 63rd Annual Meeting of the Korean Society for Cognitive and Biological Psychology, Seoul, South Korea.

3. **Yoon, H., Moon, J., & Kwon, O.-S.** (2025, February 12–14). *Attractive serial dependence occurs after motion repulsion*. [Poster session]. The 62nd Annual Meeting of the Korean Society for Cognitive and Biological Psychology, Gwangju, South Korea.
4. **Moon, J., Yoon, H., & Kwon, O.-S.** (2024, February 1–3). *Probing bidirectional serial dependence in an N-back orientation estimation task*. [Talk session]. The 61st Annual Meeting of the Korean Society for Cognitive and Biological Psychology, Seoul, South Korea.
5. **Moon, J., Saftari, L. N., & Kwon, O.-S.** (2023, February 16–17). *An environmental motion presented ahead of body motion modulates the causal inference in heading direction estimation*. [Talk session]. The 60th Annual Meeting of the Korean Society for Cognitive and Biological Psychology, Seoul, South Korea.
6. **Moon, J., & Kwon, O.-S.** (2021, July 15–16). *The effects of behavioral variability and biases on confidence in perceptual estimation*. [Talk session]. The 58th Annual Meeting of the Korean Society for Cognitive and Biological Psychology, virtual.
7. **Moon, J., & Kwon, O.-S.** (2020, August 27–28). *A key role of non-directional orientation in the coding of visual motion direction*. [Talk session]. The 57th Annual Meeting of the Korean Society for Cognitive and Biological Psychology, virtual.
8. **Moon, J., & Kwon, O.-S.** (2019, February 14–15). *Sequential effects in motion direction estimation*. [Talk session]. The 56th Annual Meeting of the Korean Society for Cognitive and Biological Psychology, Pyeongchang, South Korea.
9. **Choe, S., Moon, J., & Kwon, O.-S.** (2018, January 25–26). *Temporal dynamics of visual attention*. [Poster session]. The 55th Annual Meeting of the Korean Society for Cognitive and Biological Psychology, Suwon, South Korea.

## INVITED TALKS

|      |                                   |                                                      |
|------|-----------------------------------|------------------------------------------------------|
| 2026 | PAL Lab (PI: Oh-Sang Kwon)        | Ulsan National Institute of Science and Technology   |
| 2026 | The Neuroscience Dialogues        | Department of Neuroscience, UT Austin                |
| 2025 | UT STEM Seminar                   | Korean-American Scientists and Engineers Association |
| 2024 | Goris Lab (PI: Robbe L. T. Goris) | The University of Texas at Austin                    |
| 2024 | Kiani Lab (PI: Roozbeh Kiani)     | New York University                                  |

## HONORS AND AWARDS

|              |                                 |                                                        |
|--------------|---------------------------------|--------------------------------------------------------|
| 2026         | PCN Travel Award (€500)         | German Primate Center                                  |
| 2024         | Data Challenge Award            | Korean Society for Cognitive and Biological Psychology |
| 2022         | CSHL Course Award (\$1,500)     | Howard Hughes Medical Institute                        |
| 2021, 22, 24 | Best Oral Presentation Award    | Korean Society for Cognitive and Biological Psychology |
| 2018 – 2024  | UNIST Graduate Scholarship      | Ulsan National Institute of Science and Technology     |
| 2016         | Samsung Dream Class Scholarship | Samsung Welfare Foundation                             |
| 2012         | UNIST Dream Scholarship         | Ulsan National Institute of Science and Technology     |
| 2012 – 2018  | UNIST Undergraduate Scholarship | Ulsan National Institute of Science and Technology     |

## SUMMER SCHOOLS

|      |                                    |                               |
|------|------------------------------------|-------------------------------|
| 2026 | Primate Cognitive Neuroscience     | German Primate Center         |
| 2022 | Computational Neuroscience: Vision | Cold Spring Harbor Laboratory |
| 2021 | Deep Learning                      | Neuromatch Academy            |

## TEACHING

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### Guest Lectures

|      |                                          |                                                   |
|------|------------------------------------------|---------------------------------------------------|
| 2026 | Neural and Computational Basis of Vision | Depts. of Psychology and Neuroscience, UT Austin  |
| 2025 | INP Bootcamp                             | Interdisciplinary Neuroscience Program, UT Austin |

### Teaching Assistant

|              |                           |                                             |
|--------------|---------------------------|---------------------------------------------|
| 2019, 20, 23 | Experimental Design       | Department of Biomedical Engineering, UNIST |
| 2019, 20, 22 | Sensation and Perception  | Department of Biomedical Engineering, UNIST |
| 2017 – 2018  | Science of Human Behavior | School of Liberal Arts, UNIST               |

### Educational Outreach

|      |                                   |                                            |
|------|-----------------------------------|--------------------------------------------|
| 2016 | World Friends Korea IT (in Ghana) | National Information Society Agency        |
| 2016 | Samsung Dream Class               | Samsung Welfare Foundation                 |
| 2014 | Educational Outreach Program      | ROK Air Force Education & Training Command |

## MENTORING

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### UT Austin

|        |                           |                               |
|--------|---------------------------|-------------------------------|
| 2024 – | Tien Dung Nguyen          | Ph.D. student in Neuroscience |
| 2024 – | Avinash Ranjan            | Ph.D. student in Neuroscience |
| 2024 – | Akash Raj Gunniya Prakash | Ph.D. student in Psychology   |

### UNIST

|             |             |                                        |
|-------------|-------------|----------------------------------------|
| 2023 – 2024 | Hoyeon Yoon | M.S. student in Biomedical Engineering |
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## SERVICE

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### Societal Service

|             |                                    |                         |
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| 2026 – 2028 | Student-Postdoc Advisory Committee | Vision Sciences Society |
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### Departmental Service

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| 2024 – | CPS Colloquium Series Coordinator | Center for Perceptual Systems, UT Austin |
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### Ad Hoc Journal Reviewer (\*formal co-review)

BMC Biology; eLife; Journal of Experimental Psychology: Human Perception and Performance; Journal of Vision; Nature Communications\*; Nature Human Behaviour\*; Nature Reviews Neuroscience\*; Psychological Research; Science\*; STAR Protocols

### Ad Hoc Conference Reviewer

Cognitive Computational Neuroscience (CCN)

## REFERENCES

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Robbe L. T. Goris, Ph.D.  
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Center for Perceptual Systems  
The University of Texas at Austin  
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